Smart AI Fan — Technical Abstract

TECHNOMANCERS (T-2026) · Track 1 · Fan · atomQUEST 2026 · Stage 1 (Round 1 closes 18 Oct 2026, 18:29 GMT — gradpartners.in/e/atomquest-2026-339)

A · Team & Institutional Context

TECHNOMANCERS	T-2026	Aditya Institute of Technology and Management, Tekkali	Electronics and Communication Engineering (ECE)	Harsha Vardhan Malipeddi — harshavardhanmalipeddi1@gmail.com

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Five members — within the official 1–5 rule. Mentor and portal registration fields are not stated in this abstract; they will be supplied from the official portal if required.

B · Problem & Idea Framing

Track and specific areas of exploration — combining three:

Smart-home triggers (fan as enabler — ambient temperature/humidity triggers to control connected devices) · **sensor-based multi-directional swing** (pan/tilt steering of the full airflow disc) · **blade design for efficiency** (composite fan blades for higher air delivery at lower input power). Why combined: one sensing-and-control chain drives both comfort automation and swing actuation, and the blades set the operating point that both act on — a single product builds all three coherently.

Proposed idea/mechanism, in one paragraph:

The Smart AI Fan is a 1,400 mm-sweep ceiling fan whose local controller reads temperature and humidity and runs comfort automation with manual-override precedence and defined safe fallback. A self-locking worm-driven pan/tilt head steers the whole airflow disc (rotor stopped before tilt; tilt clearance-gated); proposed quasi-isotropic [0/±45/90]s carbon/epoxy blades run on a 24 V sensorless-FOC BLDC drive sized to IS 374 targets; and a locked-firmware supervisor MCU enforces plausibility gates, tilt interlocks and a raw power-cut path over the motor and actuators.

Baseline-vs-target performance claim, quantified:

Air delivery	Fixed-speed reference fan class	≥ 245 m³/min (IS 374 target; ≈ 8.7 m³/min/W indicative at 28.3 W)	Target; not measured
Comfort automation	Manual switching only	Automated response ≤ 5 s; ≥ 95 % success over 10–20 defined scenarios	Acceptance target
Airflow direction	Fixed direction	Steerable airflow ±20° tilt, clearance-gated	Target; gated by geometry
Smart-home trigger	None (no sensing)	Connected-device trigger path (e.g. exhaust on humidity ≥ 65 %RH)	Demonstrated on safe test load
Safety fallback	None defined	Defined safe fallback on sensor disconnect or network loss	Acceptance target

No measured baseline of ours exists; no competitor numbers are invented — every target above is stated as a target and is traceable to the evidence repository (Section D).

Any other technical detail (B): safety scope and boundaries are stated under C; remaining risks and unresolved items are preserved under D and in the repository's risk register.

C · Technical Feasibility

How the product works and why it is technically feasible — core mechanism, key components and subsystems, and safety-critical handling:

Sensors feed comfort automation and closed-loop sensorless-FOC speed control (settle target ≤ 0.5 s). Plausibility gates (–10…+80 °C, ≤ 3 K/s) route implausible sensor input to a safe default curve; a hardware NTC trips at 165 °C; the supervisor enforces a 350 ms watchdog and a raw power-cut path. The self-locking worm holds pan/tilt position on power loss — backed by absolute encoders, limit switches and hard stops, so tilt failure needs three simultaneous defenses to breach. Rotor stops before any tilt motion. Every subsystem maps to standard components, so a safe smart-subsystem demonstrator (sensing, comfort logic, override, fallback, device triggering on a safe test load) is feasible immediately, while rotor fabrication and powered-rotor validation stay gated by the T-series programme.

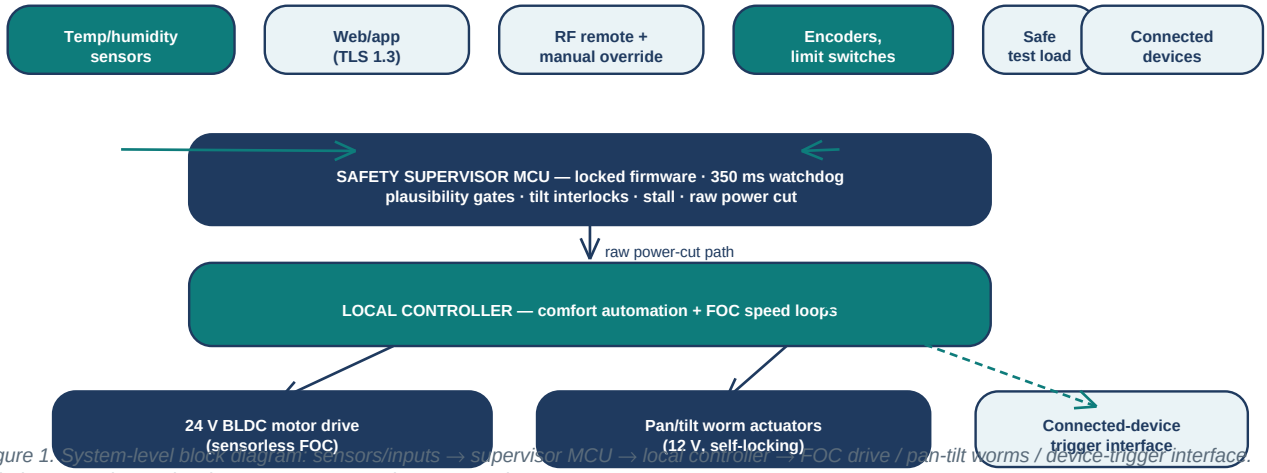

Figure 1. System-level block diagram: sensors/inputs → supervisor MCU → local controller → FOC drive / pan-tilt worms / device-trigger interface. Priority: supervisor → local → RF remote → web → automation.

Relevant numerical specifications (all values traceable to the repository; full table in the annexure):

Air delivery	>= 245 m3/min	Design target; not measured
Service value	>= 4.1 m3/min/W required; ~8.7 indicative at 28.3 W assumed input	Indicative calculation only
Blade-tip speed cap	<= 30 m/s	Standard requirement
Tip speed at 380 RPM	27.86 m/s	Calculated
Top speed	380 RPM	Target; not measured
456 RPM case	Hazard case only (tip 33.4 m/s)	Hazard case; never an operating target
Rotor inertia	~0.113 kg*m2	Assumption pending CAD sign-off
Rotor energy @380 RPM	~89.5 J	Calculated from target masses
Rotor energy @456 RPM	~128.8 J	Calculated; hazard case
Blade containment rating basis	130 J	Rating basis BLOCKED pending authorization
Moving mass roll-up	7.00 kg (sum) vs 6.80 kg (earlier stated)	UNRECONCILED; both retained
Blade mass	165 g/blade	Target; assumptions pending T-01/T-05
Blade material	quasi-isotropic [0/±45/90]s carbon/epoxy ~1.55 g/cc ~70 GPa	Assumption; supplier data required
Motor torque	>= 0.62 Nm continuous; >= 2.5 Nm 5 s peak	Specified on paper

Continued in the annexure PDF: full specification table, rotor/containment status (128.8 J vs 130 J — thin margin, BLOCKED), anchor/suspension factors, clearance derivation, T-00…T-21 validation gates.

D · Execution Feasibility

Rough list of components for the product:

3-blade CFRP composite rotor set (custom prepreg; tooling share)	6,450	CUSTOM + LONG LEAD
24 V sensorless-FOC BLDC motor (>= 0.62 Nm cont / 2.5 Nm 5 s peak)	3,700	OFF-THE-SHELF
FOC drive / controller board	1,900	OFF-THE-SHELF
Supervisor MCU board (locked firmware)	1,400	OFF-THE-SHELF
Local controller MCU board	1,150	OFF-THE-SHELF
Temp/humidity sensor set	850	OFF-THE-SHELF
Pan worm actuator (>= 2.5 Nm self-locking)	1,900	CUSTOM-RATED
Tilt worm actuator (>= 6 Nm self-locking)	2,250	CUSTOM-RATED
Absolute encoders + limit switches + hard stops	950	OFF-THE-SHELF
24 V SMPS power supply	1,000	OFF-THE-SHELF
Wi-Fi/BLE connectivity module (optional)	750	OFF-THE-SHELF
RF remote + manual override interface	700	OFF-THE-SHELF
Connected-device trigger interface or safe test load	850	OFF-THE-SHELF
Test instrumentation (power meter + reference sensors)	1,100	OFF-THE-SHELF
SS304 safety cable + anchor hardware	800	OFF-THE-SHELF
Custom mounting and pan/tilt bracket plates	900	CUSTOM
BOM total per fan (order of magnitude; no quotations — estimates only)	**26,650**	

python calculations/fan_design_calculations.py recomputes and asserts this exact roll-up. 3-fan programme ≈ INR 5.8 lakh (3 × BOM + 10 % contingency + non-recurring programme items), planning level.

System-level block diagram:

Figure 1 above (drawn directly from docs/img/system-block-diagram.svg in the repository).

Component sourcing risk — custom-fabricated, imported or long-lead:

- **CFRP blades — CUSTOM + LONG LEAD:** [0/±45/90]s prepreg needs mould tooling, autoclave/oven cure and property verification; no supplier datasheet on file; strength and mass remain assumptions.
- **Worm actuators — CUSTOM-RATED:** ≥ 6 N·m tilt / ≥ 2.5 N·m pan with self-locking at rated torque; supplier confirmation required (off-the-shelf units may not meet both).
- **Safety-relevant parts:** supervisor power-cut path, limit switches, cable/anchors — ratings must be confirmed; compliance (hipot/leakage/EMC) must be verified by accredited testing regardless of supplier claims.

Build plan to Round 2 — abstract to a working Level 1 prototype before the next deadline:

- **M1:** low-voltage sensing + comfort control on a safe test bench (sensors, controller, safe load).
- **M2:** verify override precedence (supervisor → local → RF → web → automation) and safe fallback on sensor disconnect and network loss.
- **M3:** demonstrate the connected-device trigger on a safe test load.
- **M4:** record results against predefined acceptance tests (16-repo scenario set, accuracy, response time) — then, only after independent safety review and formal approval, motor/rotor integration for Round 2.

Evidence & calculation repository — every figure in this abstract is checkable there

https://github.com/harsha-vardhan-2006/atomquest — contains: all design calculations with formulas and assumptions (calculations/fan_design_calculations.py, self-checking), the Round 1 scenario framework and controller simulation with 16 acceptance scenarios (check_scenarios.py, data/scenarios.csv), the full specification table (data/design_targets.csv), rough BOM (data/bom.csv), validation gates T-00…T-21 (data/validation_gates.csv), the preserved risk register (data/risk_register.csv) and the formulas document (docs/formulas.md); includes both submission PDF generators.

Any other technical detail (D):

This is a paper design — no hardware, nothing measured; targets remain targets. Round 1 covers sensing, comfort logic, override, safe fallback and device triggering on a safe test setup. Nothing here authorizes fabrication, energization, powered-rotor or overspeed testing. Unresolved items are preserved, not hidden — mass roll-up 7.00 vs 6.80 kg; containment 128.8 J vs 130 J; CFRP properties; electrical/EMC compliance (risk register, closed gate-by-gate T-00…T-21).